Hyperledger Sawtooth Architecture: An Introduction to the Validator Network

This article is for open source technology enthusiasts who are interested in blockchain technologies and have a working knowledge of the Hyperledger Sawtooth project. It will give them a quick overview of the network layer in the Hyperledger Sawtooth architecture.

The network layer is responsible for communication between validators in a Sawtooth network, including performing initial connectivity, peer discovery and message handling.

- Upon start-up, validator instances begin listening on a specified interface and port for incoming connections.
- Upon connection and peering, validators exchange messages with each other based on the rules of a gossip or epidemic protocol.

A primary design goal is to keep the network layer as self-contained as possible — the application should not need to understand implementation details of the network in order to send and receive messages.

Services

Sawtooth has adopted the 0MQ (Zero MQ) asynchronous client/server pattern. This consists of a 0MQ *router* socket on the server side which listens on a provided endpoint, with a number of connected 0MQ *dealer* sockets as the connected clients.

The 0MQ guide describes the features of this pattern as follows:

- Clients connect to the server and send requests.
- For each request, the server sends 0 or more replies.
- Clients can send multiple requests without waiting for a reply.
- Servers can send multiple replies without waiting for new requests.

States

Sawtooth defines three states related to the connection between any two validator nodes.

- Unconnected
- Connected - A connection is a required prerequisite for peering.
- Peered - A bi-directional relationship that forms the base case for application-level message passing (gossip).

Message types

This protocol includes the following types of messages.